



moderate/medium, and low. We mapped {Low, High and Moderate} levels into discrete form as $1 \leq Y \leq 3$ to be used in proposed simulation model. As an example: Priority level “high” is associated to **Gunshot, Perforation, Blunt trauma** and thus given discrete value 1 while “moderate” level surgeries which might include **Acute and perforated appendicitis, strangulated hernia** are given value 2. Finally, the lowest critical surgeries **Abscess, Excision of lipoma**) is assigned priority 3.

The model for generating random priorities between 1 and 3 is given in Eq (1),

$$x_i = (A * x_{i-1} + C) \bmod M \quad (1A)$$

$$Y = (b - a)x_i + a \quad (1B)$$

where A, C and N are suitably chosen constant. One set of numbers that works well are: $A = 55$, $C = 9$, $M = 1994$. Multiplying x_{i-1} by A and adding C forms large value, and the modulo operation brings the value into the range from 0 to M-1. The process begins with some “seed” value chosen for x_0 . Because the numbers only give an appearance of randomness, they are called pseudo random numbers.

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EQ (1A) represents simple's model seed based model renowned as Linear Congruential Generator (LCG). The generation of discrete RV $1 \leq Y \leq 3$ is generally a two-step process where first LCG provides numbers between 0 and 1 i.e. $X \sim U[0,1]$ and these uniformly distributed numbers are scaled and shift to meet the bounds of desired random variable as show in EQ (1b)

4.2 Patient Arrival: Poisson distribution

Generally, the patient arrival process is treated as counting and birth process in literature and thus follows a discrete Poisson distribution. Arrival distribution is the heart of queuing model of simulation. In our simulation model, this distribution represents the patients' arrival into the hospital's emergency (trauma) department. The Poisson model in EQ(2) is helpful in generating expected number of arrivals which is the only required parameter and the same is used to as mean number of arrivals, variance, and standard deviation to simulate the arrival distribution as shown in EQ(3), EQ(4) and EQ(5),

$$P(N_t = n) = \frac{e^{-\lambda t} (\lambda t)^n}{n!} \quad (2)$$

Where

N_t = Number of patient arrivals in a specified time with $N_0 = 0$

λ = Expected number of arrival events

	A	B	C	D	E	F	G	H	I	J
1	A	55		LCG	$Y=(b-a)x_i+a$					
2	M	1994		$R=(A*Z_{i-1} +C)\text{mod } M$	$1 \leq Y \leq 3$		x_i			
3	Zo	1E+07			$a=1, b=3^{x_{i-1}}$					$D7*(3-1$
4	C	9					0.333501			
5										
6	Simulation	z (initial seed)	R (LCG)x1	Random number	Generate priority					
7	1	10112166	665	0.333500502	2					
8	2	665	692	0.347041123	2		FALSE			
9	3	692	183	0.091775326	1		0			
10	4	183	104	0.052156469	1		0			
11	5	104	1741	0.873119358	3		FALSE			
12	6	1741	52	0.026078235	1		2			
13	7	52	875	0.438816449	2		FALSE			
14	8	875	278	0.139418255	1		2			

D7 =C7/\$B\$2

	A	B	C	D	E	F	G	H	I
1	A	55		LCG	$Y=(b-a)x_i+a$				
2	M	1994		$R=(A*Z_{i-1} + C)\text{mod } M$	$1 \leq Y \leq 3$		x_i		
3	Zo	10112166			$a=1, b=3^{x_{i-1}}$				
4	C	9					0.333501		
5									
6	Simulation	z (initial seed)	R (LCG)xi	Random number	Generate priority				
7	1	10112166	665	0.333500502	2				
8	2	665	692	0.347041123	2		FALSE		
9	3	692	183	0.091775326	1		0		
10	4	183	104	0.052156469	1		0		
11	5	104	1741	0.873119358	3		FALSE		
12	6	1741	52	0.026078235	1		2		
13	7	52	875	0.438816449	2		FALSE		
14	8	875	278	0.139418255	1		3		

Sheet1

Sheet2



E7 =D7*(3-1)+1

	A	B	C	D	E	F	G	H	I
4	C		9						
5							0.333501		
6	Simulation	z (initial seed)	R (LCG)xi	Random number	Generate priority				
7	1	10112166	665	0.333500502	2				
8	2	665	692	0.347041123	2		FALSE		
9	3	692	183	0.091775326	1		0		
10	4	183	104	0.052156469	1		0		
11	5	104	1741	0.873119358	3		FALSE		
12	6	1741	52	0.026078235	1		2		
13	7	52	875	0.438816449	2		FALSE		
14	8	875	278	0.139418255	1		3		
15	9	278	1341	0.672517553	2		FALSE		
16	10	1341	1980	0.992978937	3		FALSE		
17	11	1980	1233	0.618355065	2		FALSE		

Sheet1

Sheet2

